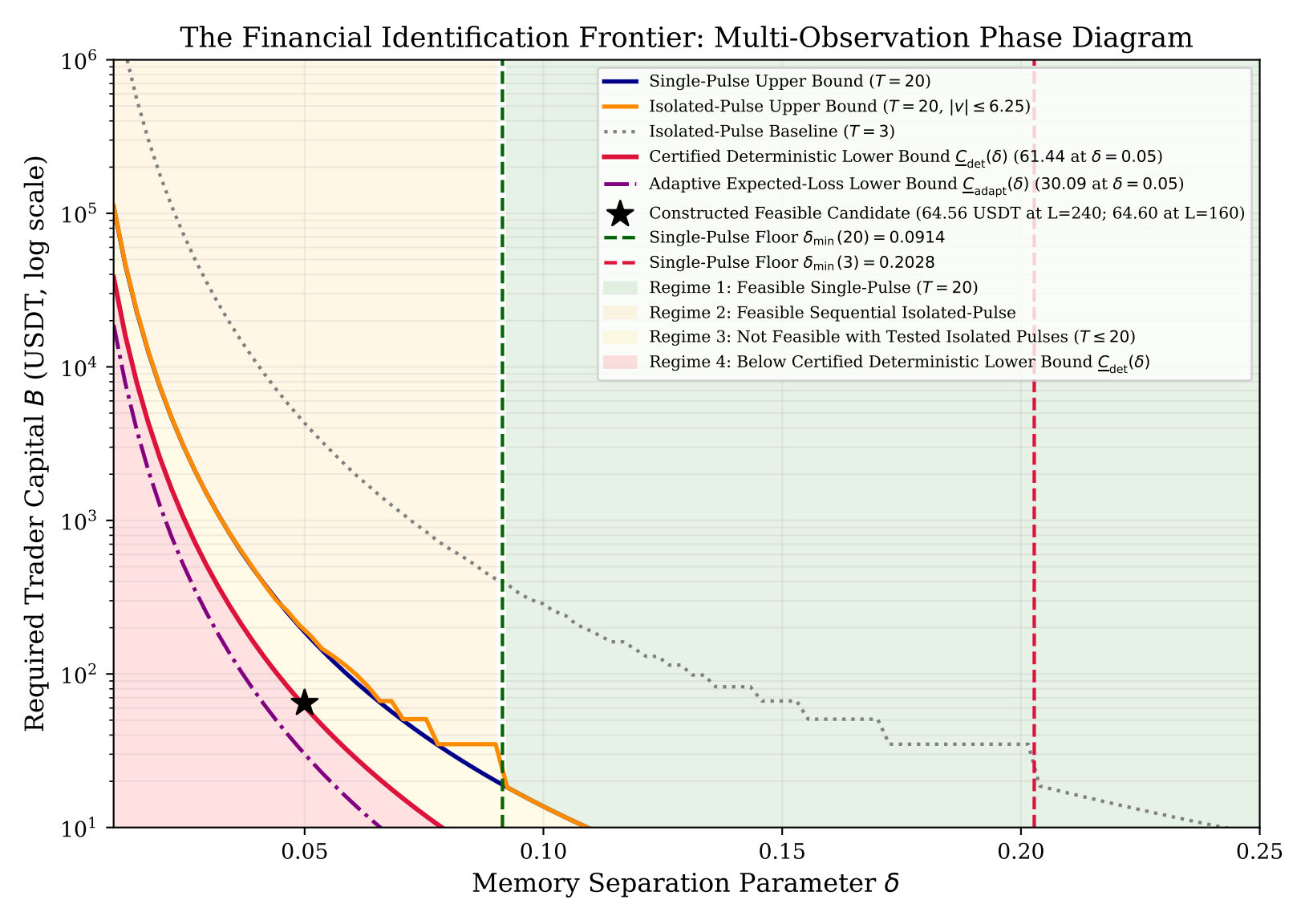
The Financial Identification Frontier: Multi-Observation Phase Diagram
Required Trader Capital B (USDT, log scale)
Memory Separation Parameter δ
10^6
10^5
10^4
10^3
10^2
10^1
0.05
0.10
0.15
0.20
0.25
Single-Pulse Upper Bound ($T = 20$)
Isolated-Pulse Upper Bound ($T = 20$, $|v| \le 6.25$)
Isolated-Pulse Baseline ($T = 3$)
Certified Deterministic Lower Bound $\underline{C}_{\text{det}}(\delta)$ (61.44 at $\delta = 0.05$)
Adaptive Expected-Loss Lower Bound $\underline{C}_{\text{adapt}}(\delta)$ (30.09 at $\delta = 0.05$)
Constructed Feasible Candidate (64.56 USDT at L=240; 64.60 at L=160)
Single-Pulse Floor $\delta_{\min}(20) = 0.0914$
Single-Pulse Floor $\delta_{\min}(3) = 0.2028$
Regime 1: Feasible Single-Pulse ($T = 20$)
Regime 2: Feasible Sequential Isolated-Pulse
Regime 3: Not Feasible with Tested Isolated Pulses ($T \le 20$)
Regime 4: Below Certified Deterministic Lower Bound $\underline{C}_{\text{det}}(\delta)$